

Exploring Public Budgets with Symber Tree: An Interactive Hierarchical Visualization for Comparisons Across Orders of Magnitude

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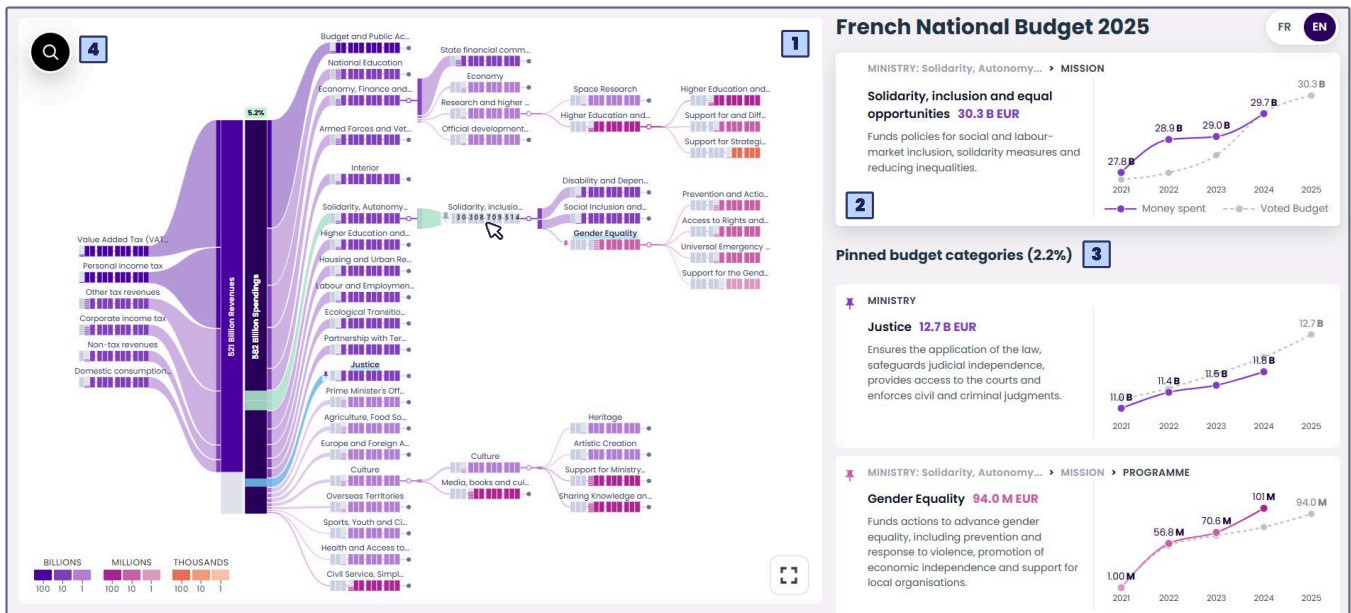


Figure 1: Interactive hierarchical visualization of the French national budget 2025, showing revenues and expenditures with Sankey-like flows and node glyphs (1), and supporting details-on-demand (2), pinning (3), and search-based navigation (4).

Abstract

We present *Symber Tree*, a novel hierarchical visualization designed to support exploration and comparisons across values spanning multiple orders of magnitude. Common hierarchical visualizations, such as treemaps and icicle plots, struggle to represent such disparities across hierarchical levels and rely on interaction techniques that increase cognitive load by requiring users to recall previously seen values. To address these limitations, we adopt a hybrid node-link approach that encodes values along both edges and nodes. A Sankey-like representation conveys linear differences between categories, while glyphs inspired by symbolic number representation support cross-scale comparison and provide reference points on

demand to facilitate evaluative meaning. We demonstrate the approach using the French national budget and report findings from a preliminary heuristic evaluation with eight visualization experts.

CCS Concepts

• **Human-centered computing** → **Graph drawings**; **User studies**.

Keywords

hierarchies, glyphs, orders of magnitude, budget visualization



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1 Introduction

In this work, we present the design and a preliminary evaluation of an interactive hierarchical visualization that aims to support quantitative understanding of public budget data. Increasingly, national governments around the world make their budgets open to the public [17, 18] in an effort to promote transparency, foster public trust, and support democratic dialogue within civil society. However, public availability does not guarantee public understanding. National budgets are inherently complex: they are typically hierarchically organized, distributing each ministry’s spending across multiple subcategories, and they contain numerical values spanning several orders of magnitude—from thousands to millions and billions. We refer to such values as *Orders of Magnitude Values (OMVs)*.

People often struggle to reason about OMVs, particularly when these values extend far beyond the limits of everyday experience [5, 12]. Research in cognitive psychology emphasizes that quantitative understanding of large and abstract numbers depends on *evaluative meaning*—that is, the ability to interpret a value relative to meaningful reference points [19]. For example, whether one billion euros constitutes a large or small amount depends on comparisons with other allocations or with past budgets. Supporting contextual and comparative reasoning is therefore essential for enabling quantitative understanding of budget figures across a wide range of stakeholders.

However, dominant hierarchical visualization techniques, such as treemaps or icicle plots, are poorly suited for supporting quantitative comparisons between nodes that differ greatly in order of magnitude [6]. Large nodes tend to dominate the display, while smaller yet potentially important categories become difficult to see or compare. Interaction techniques such as semantic or geometric zooming can reallocate space and reveal additional detail, but they typically rely on multiple views and sequential navigation [4]. As a result, users must mentally recall previously seen values, increasing cognitive load and hindering direct comparison.

To address these limitations, we present **Symber Tree**, a novel hierarchical visualization inspired by principles of **symbolic number** representation, including scientific notation (mantissa and exponent), lexical numerals (scale words), and the decimal system. Our goal is to provide an interactive visualization that (1) supports quantitative comparisons across hierarchical levels and orders of magnitude, and (2) offers reference points on demand to facilitate the evaluative interpretation of abstract budget figures. We focus on the case of the French national budget, as it is openly available and well documented, and present the rationale behind our design and interaction choices, a high-fidelity interactive prototype, and the results of a preliminary heuristic evaluation conducted with eight visualization experts.

2 Design Considerations

In the following sections, we detail the visualization design and interaction choices underlying Symber Tree.

2.1 Visualization

We use a hybrid node–link tree representation [16] as the basis of our visualization. Node–link trees are a widely used and familiar representation of hierarchical structures, reflecting conventions

established in domains such as genealogy, file systems, and biological taxonomy [22]. Although they have been criticized for their suboptimal use of space compared to space-filling techniques such as treemaps and icicle plots, interactive exploration methods based on degree of interest (DOI) functions [7] and direct manipulation can mitigate these limitations [9, 20]. For example, dynamically expanding or collapsing branches supports selective exploration by allocating space on demand to areas of interest, which is particularly beneficial in the budget context for supporting comparisons across hierarchical levels within a single view. However, while node–link trees are effective for navigating and understanding hierarchical structures, they are not inherently designed to support quantitative comparisons between nodes. To address this limitation, we adopt a hybrid visualization approach [16], retaining the node–link layout for navigation while redundantly encoding values in both edges and nodes. Specifically, we use a Sankey-like representation [21] along the edges to convey money flow and linear differences, and we design a custom glyph for each node to support OMV representation and quantitative comparison across the tree. We deliberately decided not to use DOI functions for tree exploration, allowing users to explicitly control node visibility.

Sankey Diagram for Representing Budget Distributions on a Linear Scale. Sankey diagrams are commonly used to represent flows—such as energy, materials, or money—between entities in networks and processes [21]. They leverage link width as a visual metaphor for quantity, supporting understanding of how resources are distributed and transformed. In the context of public budget data, this metaphor naturally aligns with the notion of money flowing from higher-level budget categories to increasingly specific allocations. It also supports reasoning about how incoming revenues are transformed and redistributed into expenditures, providing a coherent view of the budget as a whole. While linear flow encodings in Sankey diagrams are effective for supporting value retrieval and comparison at larger magnitudes [8], they fall short when applied to OMVs. In such cases, differences between smaller flows become visually negligible due to minimum edge thickness constraints [14]. To address this limitation, we retain the Sankey representation to convey the overall flow of money and introduce a custom glyph at each node to support value retrieval and comparison across OMVs. For each node, its children are sorted by value from bottom to top.

Glyphs for Orders of Magnitude Values. To design glyphs that effectively encode OMVs—making both large and small magnitudes visible and comparable—we draw on insights from numerical cognition [11] and visualization research [2, 3]. We round values to reduce cognitive effort [15] and decompose OMVs into three components: (1) the **mantissa**, (2) the **exponent**, following scientific notation, and (3) **scale words** (e.g., billions, millions), which reflect the role of categorical information in aiding the perception of large numbers [11]. Figure 2 illustrates our design process using numbers spanning five orders of magnitude.

- (1) We begin with the piecewise linear *EplusM* scale that outperforms other scales for visualizing OMVs [2]. Using *EplusM*, exponents are evenly spaced along the axis—similar to a logarithmic scale—while the mantissa is linearly interpolated

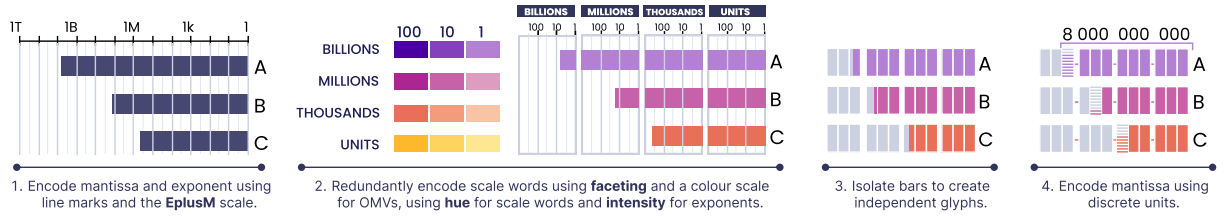


Figure 2: Glyph design with three categories differing by orders of magnitude: A = 8,000,000,000; B = 20,000,000; C = 600,000.

between adjacent exponents. Bars are right-aligned to reflect how numbers are typically represented symbolically, with higher-magnitude categories appearing to the left of lower-magnitude ones.

- (2) We incorporate scale words [3] using a redundant encoding strategy: faceting by column—aligned with how people group digits in threes for readability (e.g., using commas, as in 200,000,000)—and color (see section 2.1), to enhance the pre-attentive perception of different magnitudes in overview while maintaining the ordinal perception of scale words.
- (3) We remove the axis and treat each bar as an independent glyph. The combination of scale-word faceting and EplusM grid lines forms groups of rectangles, each representing a distinct exponent. This design serves as a visual metaphor for digit grouping, preserving the familiar structure of the decimal system. To ensure a consistent reference frame across glyphs, we allocate a fixed length based on the largest scale word and use gray to indicate missing exponents.
- (4) Finally, we divide the mantissa into discrete units and vertically stack them at the position corresponding to the exponent of the value. Colored units encode the value of the mantissa, while gray units represent the number of missing steps to the next exponent, supporting a form of subitizing [10]. These design choices enable a more accurate and discriminable encoding of the mantissa, facilitating comparisons between values in the same and neighboring magnitudes.

Color scale for Orders of Magnitudes Values. To provide a clearer sense of magnitude at the overview level, we designed a discrete color scale in which one hue is assigned to each exponent, covering a total of 12 exponents spanning values from 1 to 999,999,999,999 (see Figure 2). To ensure perceptual continuity across magnitudes [2], we base the palette on the continuous *Plasma* color scale [23]. Leveraging the categorical perception of large numbers, we group exponents by magnitude category (e.g., thousands, millions, billions). We assign a discriminable hue to each category by sampling four evenly spaced anchor points along the *Plasma* scale, while deliberately avoiding the very dark and very bright extremes. Within each magnitude category, we create an ordered set of three colors corresponding to 1, 10, and 100 by slightly shifting the sampling position around each anchor (-3%, 0, +3%) and baking these shades at different opacity levels (100%, 75%, 50%) over a white background. This approach increases discriminability within categories while preserving an ordered perception of magnitude. The resulting palette consists of four magnitude groups, each containing three ordered colors, and is applied consistently

across the entire interface to reinforce magnitude cues and support coherent interpretation throughout the system.

2.2 Interaction

Symbler Tree interactions support navigation, details-on-demand, and search.

Tree Navigation. Users can interactively navigate the hierarchical tree by clicking nodes to expand or collapse the corresponding branches. As the tree expands, the view is dynamically adjusted to keep the visible structure within the viewport. As additional hierarchical levels are revealed, labels and glyphs are proportionally scaled down to preserve context while maintaining visibility of the overall structure. Users can zoom in to inspect details by scrolling or double-clicking a desired focus point, and zoom out by scrolling in the opposite direction. The view can also be panned horizontally and vertically via click-and-drag or the keyboard’s arrow keys. Finally, a reset button located at the bottom-right corner of the visualization allows users to return the view to its initial state, centered and fitted to the available space.

Details on Demand and Pinning. Users can access details on demand for each budget category by hovering over a glyph. Upon hover, the actual number is displayed, and a corresponding flow of money associated with the node is highlighted in light green, indicating the percentage of the total budget allocated to that category (see Figure 1 (1)). At the same time, an information card appears in the top-right corner of the interface, presenting a short description of the category along with the voted and consumed budgets over the past years (see Figure 1 (2)). Additionally, a pin icon appears to the left of the glyph, allowing users to pin the information card under Pinned Budget Categories (see Figure 1 (3)). This interaction supports side-by-side comparison between multiple nodes in the tree, facilitating comparative analysis across budget categories.

Search. Users can search for a specific budget category using the search component located at the top-left of the interface (see Figure 1 (4)). As users type a query, the search returns matching categories and displays their paths in the hierarchy, from the nearest ancestor (e.g., the ministry) to the matched category. Results are ordered by the depth of the matching category in the hierarchy, prioritizing higher-level categories, such as ministries, and their descendants. When users hover over a search result, the corresponding nodes in the tree dynamically expand and are highlighted, enabling interactive, category-focused navigation. Clicking on a

result keeps the selected branch open, pins the corresponding node, and centers the view on the selected category.

2.3 Interface Design, Data and Implementation

We implemented a high-fidelity web prototype of the visualization using React.js and D3.js. The interface consists of four main components (see Figure 1): (1) a hierarchical visualization showing revenues and expenditures for the selected year; (2) a dedicated panel displaying additional details about a hovered budget category, including a short description and trends in voted and consumed budgets over the past years; (3) a panel listing selected budget categories to support comparison; and (4) a search component that facilitates category-focused exploration within the tree. To showcase the design, we used data from the French national budget. France provides open access to its budget data, along with detailed documentation in editable formats [18]. We collected datasets from the past five years, covering both voted and consumed budgets, and constructed a JSON file with a predefined budget hierarchy following the structure used in official government documents. The French general budget is organized hierarchically across multiple levels: **ministries, missions, programs, actions**, and, in some cases, **sub-actions**, resulting in a hierarchy of up to five levels. The dataset contains 1,085 nodes with values ranging from tens of thousands to hundreds of billions, spanning eight orders of magnitude. The JSON file containing the data, along with the code for the visualization interface, is provided as supplemental material for extension and reuse and is also deployed online via a temporary link provided for anonymization purposes: <https://french-budget-visualization.spacerok.workers.dev/> (unrelated to the authors).

3 Usability Evaluation

In this section, we present the process and preliminary results of a formative usability study with visualization experts.

3.1 Participants and Procedure

We recruited eight visualization experts (P1–8), all of whom are involved in the design, development, or research of data visualization systems. We conducted an IRB-approved formative usability evaluation following established heuristic evaluation practices [13]. The primary goal of this evaluation was to identify design and interaction issues in the visualization and interface at an early stage, in order to iteratively refine the system before conducting evaluations with domain users familiar with budgeting data, such as journalists, or with citizens with varying levels of data literacy. The study began with a brief demographic and background questionnaire to assess participants' familiarity with budgeting data, human-computer interaction, evaluation methods, and visualization. We then conducted an onboarding phase, introducing the visualization design and interaction. This was followed by a task-based evaluation, during which participants performed representative tasks (navigation, value retrieval, finding extrema, comparisons) while following a think-aloud protocol, allowing us to capture usability breakdowns, interaction difficulties, and issues related to visual interpretation. At the end of the task-based session, participants completed the ICE-T questionnaire to rate our prototype on Insight, Confidence, Essence, and Time perspectives [24]. Finally,

they took part in a semi-structured interview focused on perceived strengths of the proposed solution, identified usability problems, and concrete suggestions for design improvements. Each evaluation lasted approximately 1 hour. The specific tasks, the heuristic evaluation questionnaire, and results are provided in the supplemental materials.

3.2 Preliminary results

The preliminary evaluation yielded consistently positive feedback across all assessed dimensions, with average ICE-T scores above 6 out of 7 for insight generation, confidence, essence, and time. Participants found the node–link layout effective for understanding the hierarchy and comparing nodes across different branches, and none reported difficulties navigating the tree. The Sankey-like design supported a linear sense of magnitude differences, particularly within branches, while the color scale enabled rapid comparison across orders of magnitude at a glance. For nodes within the same order of magnitude, participants occasionally relied on zooming or hovering to access exact values, slightly increasing response time; however, they reported that overview-level similarity was sufficient, with details accessed on demand when needed. The glyph design was perceived as effective in conveying magnitude differences through redundant encodings without overwhelming numerical detail. Pinning supported deeper comparison by revealing trends over time and discrepancies between voted and consumed budgets, prompting further exploration, while search enabled discovery of semantically related categories across branches and hierarchical levels.

Identified Issues and Suggestions for Improvement. No major usability issues related to the hierarchical visualization design were identified during the evaluation. Most discussion instead focused on the reference points provided to support evaluative meaning, particularly in the interpretation of budget trends over time. Evaluators noted the need to include alerts related to data anomalies, such as missing or inconsistently reported budget entries across years. Participants also proposed supporting trend comparison across multiple scales, including a superimposed view using a common scale, as well as interactive rescaling to compare trends across orders of magnitude. Additional suggestions included enabling reorganization and grouping of pinned categories to support structured comparison, and providing percentage views relative to different ancestor levels in the hierarchy, rather than only relative to the total budget, to allow alternative framings of budget significance.

4 Conclusion and Future Work

In this work, we presented **Symber Tree**, an interactive hierarchical visualization designed to support quantitative comparison of public budget values spanning multiple orders of magnitude. Through the design of a hybrid node–link representation that combines Sankey-like flows with glyphs inspired by symbolic number representation, we aim to provide a numeracy-informed hierarchical visualization for OMVs [1]. Our goal is to better support evaluative meaning by facilitating comparison and providing multiple reference points on demand. We developed and deployed an interactive prototype of the proposed solution using the real-world example of the French national budget. A preliminary heuristic evaluation with visualization experts provided encouraging feedback on the

system's effectiveness, while also revealing issues that need to be addressed before proceeding with evaluations involving domain experts and the general public. We believe that the proposed visualization approach can be generalized to other domains involving hierarchical OMVs, such as energy or carbon emissions. In addition, components inspired by symbolic number representation—such as the color scale and glyph design—may be reused and serve as foundations for further visualization research.

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